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2025-2026 Spring Term SE 216 Software Project Management

Project Final Report

**A Centralized Platform for Educational
and Career Opportunities**



Selin TAY 20230601069

Kaan DÖNMEZ 20210601202

Doruk YAŞAR 20230602514

Ömer ŞEN 20240601071

Hilal Seray ORTAN 20240601059

Salih Batuhan ŞENER 20250601294

Baran Deniz AYDIN 20230601011

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1 - INTRODUCTION

Finding and keeping track of educational and career opportunities as a student has always been a challenge. Most of the time, relevant internships, research programs, or summer schools are either discovered too late or missed entirely. The main reason behind this is the lack of a unified system that could bring all these opportunities together in one place. Especially when opportunities are spread across dozens of different websites, university portals, and social media platforms, manually searching and tracking them is no longer a sustainable approach.

To address this issue, we developed A Centralized Platform for Educational and Career Opportunities, a smart and user-friendly application designed specifically for students and young professionals. The platform includes many useful features such as browsing personalized opportunity feeds, applying for internships and jobs, tracking application statuses, receiving deadline reminders, and managing different user roles like students, employers, and administrators.

The system also offers an AI-based recommendation engine, multi-language support, and a built-in video interview system that allows users to connect with employers directly. This way, students can discover and act on opportunities quickly and efficiently, all within a single platform.

Students, employers, and administrators can interact with the system by posting and browsing opportunities, submitting and reviewing applications, sharing past experiences through the feedback module, and receiving real-time notifications about deadlines and new listings.

The platform aims to improve the career and academic development of students while also creating a more organized, transparent, and accessible connection between opportunity providers and seekers. In this way, the system provides a more efficient, personalized, and opportunity-rich experience for every user.

2- REQUIREMENTS

2.1. Functional Requirements

1. The system shall allow users to register and log in using a valid email address and password.
2. The system shall display a personalized feed of opportunities based on the user's profile, including internships, job postings, scholarship programs, courses & certifications, summer schools and research programs.
3. The system shall provide an AI-based recommendation engine that suggests the most relevant opportunities based on the user's profile and past interactions.
4. The system shall allow users to search and filter opportunities by location, date, events, and required skills.
5. The system shall allow users to save opportunities to a personal favorites list for later access and review.
6. The system shall notify users via both email and push notifications when new matching opportunities are posted.
7. The system shall notify the user 24 hours before the deadline of items added to favorites.
8. If a training requires direct registration, the system shall prevent users from registering for another training scheduled at the same date and time.
9. The system shall allow users to share experiences and feedback about programs they attended to help others make informed decisions.
10. The system shall allow users to access reviews from people who have previously participated in the selected programs, jobs, internships, or similar activities.

2.2. Non-Functional Requirements

1. The system shall load opportunity listings within 3 seconds under normal network conditions
2. The system shall provide a consistent user experience on smartphones and tablets running Android 15 or later and iOS 17.0 or later.
3. The system shall support languages -Turkish, English, German, and Spanish- to accommodate international users and opportunities.
4. The system must comply with KVKK and GDPR to ensure user privacy.
5. The system shall encrypt all user data, including personal information, application history, and login credentials, using industry-standard encryption protocols.
6. If an application has a quota limit and the quota is full, the system shall display a 'Quota Full' message within 2 seconds. If a user attempts to register at that moment, the system shall remove the user from the registration screen and notify them within 2 seconds.
7. The system shall have an automated health-check and auto-restart script to ensure a minimum of 99% uptime.
8. The system shall implement a human verification system and rate-limiting on registration and review forms to prevent automated bots from manipulating data.
9. The system shall support at least 1000 concurrent users without response time exceeding 5 seconds.
10. The system shall comply with WCAG 2.1 accessibility standards to ensure that users with disabilities can navigate and use the platform effectively.

3 - STAKEHOLDERS

There are several key stakeholders in this project, each with specific roles and impacts on the project's success or failure:

End Users (Students and Young Professionals)

Students and young professionals are the primary users of the platform and the group most directly impacted by its outcomes. They benefit from having a single, organized hub that aggregates all types of career and educational opportunities relevant to their academic background and interests. Through the platform, they can browse a personalized feed, save opportunities to a favorites list, track their application statuses, and receive deadline reminders. Their ongoing interaction with the system also improves the quality of AI-based recommendations over time, and their feedback directly influences the continued development of the platform.

Universities and Educational Institutions

Universities and educational institutions benefit from the platform by gaining a wider channel to promote their summer schools, certificate programs, research opportunities, and Erasmus partnerships beyond their own portals. For university career offices and international relations departments, the platform reduces the burden of managing separate announcements for each program. Their cooperation in providing accurate and timely program details is critical for the reliability of the system, as outdated or incorrect listings would directly harm both the institution's reputation and the user experience.

Companies and Employers

Companies and employers are directly affected by the platform as both content providers and beneficiaries. By posting internship positions, job openings, and training programs, they gain access to a filtered audience of qualified and motivated candidates. The platform's skill-based and department-based filtering ensures that their listings reach the most relevant applicants, reducing the time and cost spent on unsuitable candidates. Their active participation is essential for keeping the opportunity feed valuable and up-to-date for students.

Platform Administrators

Platform administrators are responsible for maintaining the system's integrity, content quality, and operational reliability. They monitor new listings to verify that links are active and that posted content complies with platform guidelines. They also handle flagged content, manage user reports, and enforce compliance with KVKK and GDPR regulations. Their efficiency in responding to quality and security issues directly affects user trust and the overall health of the platform.

Development Team

The development team is responsible for designing, building, and maintaining the platform. They are directly impacted by the project's outcomes in terms of professional growth, experience, and reputation. A successful project strengthens the team's technical capabilities and serves as a strong portfolio piece, while a project failure risks wasted effort and reputational damage. Their decisions regarding architecture, tool selection, and sprint planning shape every aspect of the system's quality and sustainability.

Government and Regulatory Authorities

Government and regulatory bodies — most notably Turkey's Personal Data Protection Authority under KVKK and the European Union under GDPR — are indirect but legally binding stakeholders. The platform collects and processes significant volumes of personal data, including academic backgrounds, application histories, and behavioral interaction patterns. Compliance with the data protection requirements set by these authorities is not optional; failure to do so would expose the platform to legal penalties and operational shutdowns. Their requirements have directly shaped the system's data architecture, encryption protocols, and user consent mechanisms.

External Content Providers

External content providers include organizations, agencies, and platforms that supply opportunity listings such as bootcamp organizers, hackathon hosts, career fair coordinators, and scholarship foundations. They benefit from the increased visibility and traffic the platform brings to their programs. Their continued contribution of fresh, accurate content is what keeps the platform's feed relevant and comprehensive. Without their ongoing engagement, the platform's value to end users would diminish significantly over time.

Alumni and Experienced Users

Alumni and experienced users play a unique role in the platform's ecosystem by contributing peer reviews, experience-sharing posts, and feedback about programs they have previously attended. Their contributions help current students make more informed decisions about which opportunities to pursue. They benefit from having a structured platform where their experiences can reach and meaningfully influence a large audience of students, extending the impact of their own career journeys beyond their immediate networks.

Project Sponsors and Investors

Project sponsors and investors provide the financial backing and strategic support necessary for the platform's development and long-term sustainability. They benefit from the platform's growth in terms of financial returns and increased investment value. Their stake in the project means they have a direct interest in timely delivery, user adoption rates, and regulatory compliance. Their support also shapes the platform's roadmap priorities and scalability decisions.

Data and Statistics Organizations

Data and statistics organizations, including academic research groups and market analytics firms, represent an indirect but valuable stakeholder group. A successfully operating platform generates rich, real-world data on educational trends, career preferences, and opportunity engagement patterns among young professionals. This data holds significant research value. Their interest in the platform underscores the importance of maintaining data quality, anonymization standards, and ethical data governance throughout the system's lifecycle.

Rival and Competitive Businesses

Competing platforms and businesses — such as existing job boards, university career portals, and internship aggregator services — are affected by the platform's success in the market. Their presence shapes the competitive landscape the platform must navigate and serves as a benchmark for feature development and user experience standards. While they do not directly interact with the system, their reaction to the platform's growth, such as adding similar features or lowering prices, indirectly influences the platform's strategic direction.

Advertisers and Marketing Firms

Advertisers and marketing firms represent a potential future revenue stream for the platform. A large, engaged user base of students and young professionals is an attractive demographic for companies promoting career-related products, educational services, and professional tools. Their interest in the platform grows proportionally with its user base and engagement metrics. Careful management of this stakeholder relationship is important to ensure that advertising integration, if pursued, does not compromise the user experience or the platform's credibility as a neutral opportunity aggregator.

4 - SOFTWARE PROCESS MODEL

Software Process Name: SCRUM

Scrum is an agile project management framework emphasizing flexibility, teamwork, and iterative progress. It involves three key roles: the Product Owner, who prioritizes work and manages the backlog; the Scrum Master, who ensures the team adheres to Scrum practices and facilitates problem resolution; and the Development Team, which delivers increments of the product. Scrum employs artifacts like the Product Backlog and Sprint Backlog to track work and uses events such as Sprint Planning, Daily Scrum, Sprint Review, and Sprint Retrospective to facilitate communication and continuous improvement. The primary goal of Scrum is to deliver value to customers efficiently while promoting transparency and continuous development.

Sprint 1: Project Initiation and Planning(10 days)

- Database Planning
- Opportunity Data Setup
- User Registration and Authentication
- Role and Access Management
- API Integration Research

Sprint 2:Core Mobile Development(11 days)

- Search & Filter Functionality
- Personalized Feed System
- Application and Tryout Tracking System
- Favorites / Bookmark System
- User Profile Management

Sprint 3: AI & Reminder Systems (12 days)

- AI-Based Recommendation Engine
- Notification System
- Deadline Reminder System
- Duplicate Application Prevention
- Analytics & Reporting System

Sprint 4:Finalization, Deployment and Testing (12 days)

- Review & Feedback System
- Interview & Video Interaction System
- Multi-Language Support
- Advanced Security & Data Protection
- Testing

5 - SCHEDULE AND EFFORT

The Gantt chart and project schedule provide a detailed timeline for our project. Our schedule shows all project activities with days and dependencies, which can be easily understood by looking at the Gantt chart. The project is expected to take 43 business days to complete and is made up of 4 sprints.



ALL	INFO	TASK NAME	DURATION	PLANNED STA...	PLANNED FINIS...	PREDECESSOR
1		Sprint 1	10 days	16.02.2026	25.02.2026	
2		Database Design & Planning	4 days	16.02.2026	19.02.2026	
3		Opportunity Data Setup	3 days	20.02.2026	22.02.2026	
4		User Registration and Authentication ...	3 days	21.02.2026	23.02.2026	
5		Role and Access Management	2 days	24.02.2026	25.02.2026	4
6		API Integration Research	2 days	24.02.2026	25.02.2026	4
7		Sprint 2	8 days	26.02.2026	5.03.2026	
8		Search & Filter Functionality	4 days	26.02.2026	1.03.2026	
9		Personalized Feed System	4 days	28.02.2026	3.03.2026	
10		Application and Tryout Tracking Syst...	3 days	1.03.2026	3.03.2026	
11		Favorites / Bookmark System	2 days	2.03.2026	3.03.2026	
12		User Profile Management - 3 Days	3 days	3.03.2026	5.03.2026	
13		Sprint 3	12 days	7.03.2026	18.03.2026	
14		AI-Based Recommendation Engine - ...	5 days	7.03.2026	11.03.2026	
15		Notification System - 3 Days	3 days	12.03.2026	14.03.2026	14
16		Deadline Reminder System - 2 Days	2 days	13.03.2026	14.03.2026	
17		Duplicate Application Prevention - 2 ...	2 days	15.03.2026	16.03.2026	15,16
18		Analytics & Reporting System - 3 Days	3 days	16.03.2026	18.03.2026	
19		Sprint 4	12 days	19.03.2026	30.03.2026	
20		Review & Feedback System - 3 Days	3 days	19.03.2026	21.03.2026	
21		Interview & Video Interaction System ...	4 days	20.03.2026	23.03.2026	
22		Multi-Language Support - 2 Days	2 days	21.03.2026	22.03.2026	
23		Advanced Security & Data Protection ...	4 days	22.03.2026	25.03.2026	
24		Testing - 5 Days	5 days	26.03.2026	30.03.2026	23
25						

6 - PROJECT RISKS

This project aims to develop a centralized platform that provides students and young professionals with personalized access to opportunities such as internships, job postings, scholarship programs, courses & certifications, summer schools, and research programs. Since the system includes AI-based recommendations, notifications, and user data management, the project may face technical, security, performance, and privacy risks. Therefore, identifying and managing these risks is important to ensure system reliability and user satisfaction.

- 1. Inconsistent Data Sources :** Data from different sources may be incompatible or inconsistent, resulting in corruption of the database and causing system crashes unless handled..
- 2. Opportunity Normalization Failure :** Data cannot be standardized into a common usable format, making it impossible for users to filter or search for entries accurately.
- 3. AI Recommendation Inaccuracy:** The AI system suggests irrelevant opportunities, which quickly frustrates users and reduces the core personalization value of the platform.
- 4. Duplicate Detection Complexity:** Same opportunity appears differently across sources, making deduplication difficult.
- 5. Performance & Scalability Issues:** System slows down under high user load, causing long page load times and driving users away during peak application seasons.
- 6. Incorrect Opportunity Matching:** System may incorrectly merge different opportunities, causing users to miss distinct openings and creating general confusion.

- 7. Notification System Failure:** Users do not receive critical deadline alerts, causing them to miss important career opportunities and breaking system trust.
- 8. Review System Abuse:** Users may submit misleading or false reviews, damaging the platform's credibility as a reliable and objective resource.
- 9. User Profile Misconfiguration:** Incorrect or incomplete user input leads to poor recommendation quality, rendering the AI matching algorithm ineffective.
- 10. Data Privacy & Security Breach:** Leakage of user data leads to trust issues and severe legal penalties under data protection regulations.
- 11. Duplicate Listings:** Same opportunity appears multiple times, which clutters the user interface and significantly degrades the overall user experience.
- 12. Auto Health-check Failure:** System uptime drops due to failure in monitoring and auto-restart mechanisms, leading to prolonged and unaddressed platform downtime.

In conclusion, this project has several technical and operational risks related to data integration, AI recommendations, system performance, and user security. By identifying and analyzing these risks early, the team can develop effective prevention and management strategies to improve system reliability, user experience, and overall project success. Proper risk management will help ensure that the platform operates efficiently, securely, and sustainably for its users.

7 - SOFTWARE TOOLS

In the development of the system, various software tools were evaluated and selected for the development, database management and UI/UX design tasks of the project; taking into consideration the cost, training days and functionality.

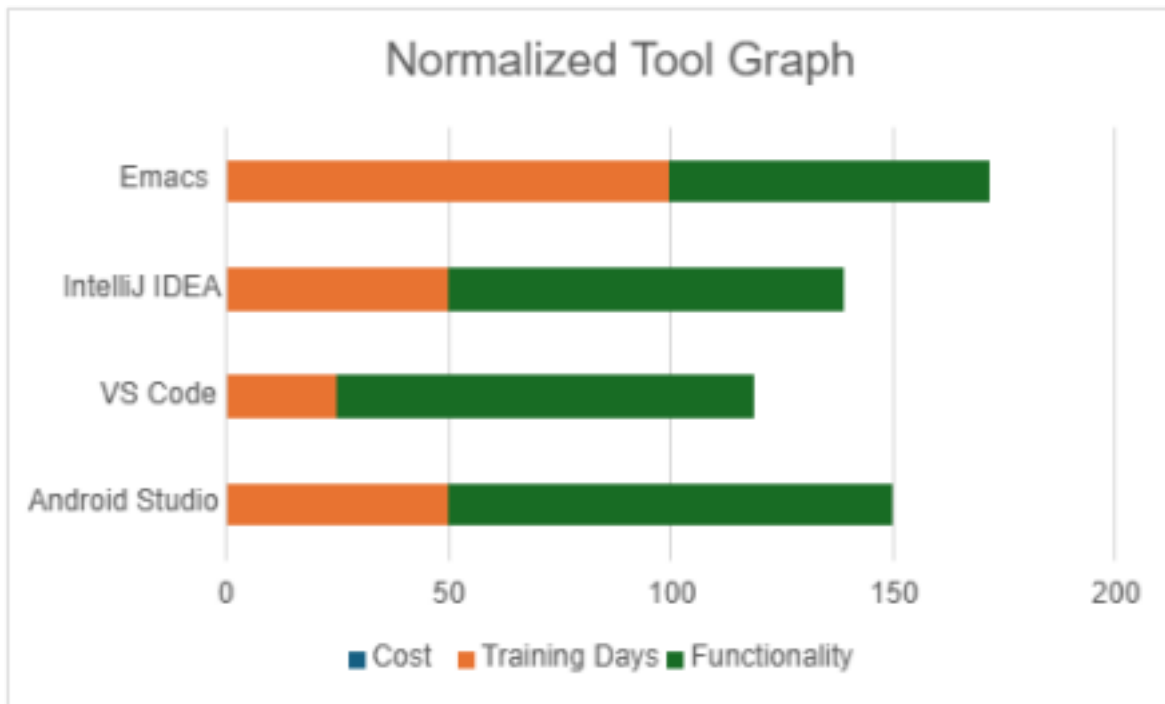
Task 1: Development of the Mobile App Using Flutter

Tool Cost/Training/Functionality Data

Tools	Android Studio	VS Code	IntelliJ IDEA	Emacs
Cost	Free	Free	Free	Free
Training Days	10	5	10	20
Functionality	90	85	80	65

Normalized Cost/Training/Functionality Data

Tools	Android Studio	VS Code	IntelliJ IDEA	Emacs
Cost	0	0	0	0
Training Days	76.02	38.46	76.92	100
Functionality	100	94.44	88.89	66.67



We chose Android Studio as our primary development environment for Flutter because it offers the most comprehensive and Flutter-specific toolset among all evaluated options. As the official Google-supported IDE for Flutter development, Android Studio provides built-in emulators, advanced debugging tools, and seamless Flutter plugin integration, making it the most suitable choice for our project.

Although VS Code has the shortest learning curve at just 5 days and offers strong Flutter support through extensions, it lacks the depth of built-in mobile development features that Android Studio provides natively. IntelliJ IDEA shares the same core engine as Android Studio but does not include Flutter-specific tools out of the box, requiring additional configuration and resulting in a lower functionality score. Cursor is the easiest to learn at only 4 days and offers powerful AI-assisted code completion, however it is primarily a general-purpose editor and does not provide the dedicated mobile development environment our project requires. Emacs, while highly customizable, has the longest learning curve at 13 days and the lowest functionality score among all options, making it impractical for a time-sensitive Flutter project.

Ultimately, Android Studio's superior functionality score, deep Flutter integration, and strong developer community support make it the most appropriate and reliable choice for developing our mobile application.

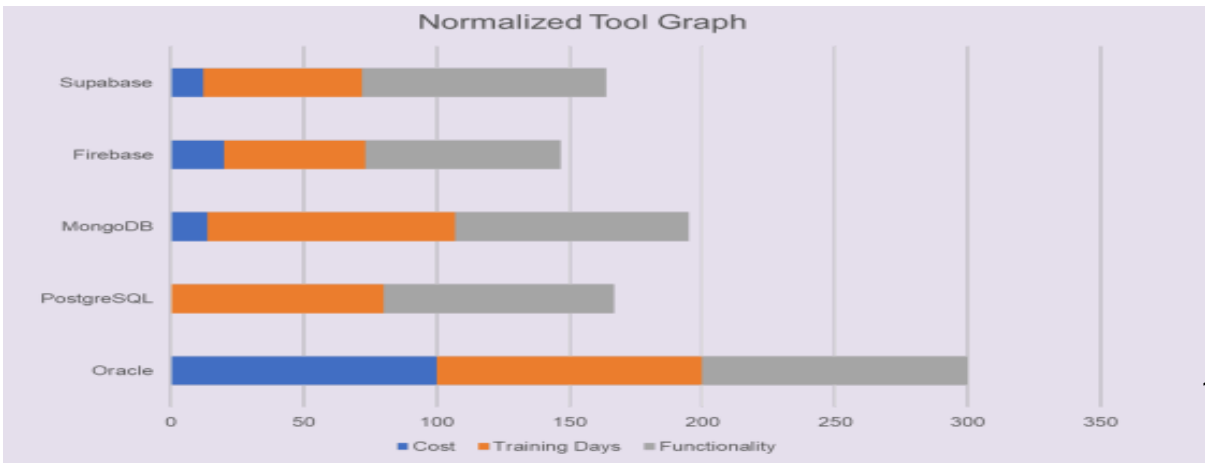
Task 2: Database Management System:

Tool Cost/Training/Functionality Data

Tools	Oracle	PostgreSQL	MongoDB	Firebase	Supabase
Cost	\$5,000/y	Free	\$684/y	\$1,000/y	\$600/y
Training Days	15	12	14	8	9
Functionality	98	85	86	72	90

Normalized Cost/Training/Functionality Data

Tools	MySQL	PostgreSQL	MongoDB	Firebase	Supabase
Cost	100	0	13.7	20	12
Training Days	100	80	93.3	53.3	60
Functionality	100	86.7	87.8	73.5	91.8



As illustrated in the comparison graph above, our selection criteria prioritized establishing a balance between high-level functionality and professional training investment. This balance is particularly important considering that all enterprise database solutions involve significantly high costs. Although PostgreSQL requires a 12-day training period and has a normalized score of 80.0, it stands out as the most sustainable solution because it offers nearly enterprise-level functionality at zero cost, compared to tools such as Supabase at \$600 and Oracle at \$5,000. From the company’s perspective, the most cost-effective choice is clearly evident.

This tool possesses the strong relational integrity and advanced security protocols necessary to ensure compliance with KVKK and GDPR, as stated in Non-Functional Requirement 4. Furthermore, it provides the superior data management capabilities required for our AI-based recommendation engine, as specified in Functional Requirement 3, and guarantees high reliability for at least 1,000 concurrent users, as stated in Non-Functional Requirement 9.

Ultimately, PostgreSQL clearly differentiates itself from other database tools through the functional depth and professional stability it provides without imposing any financial burden .

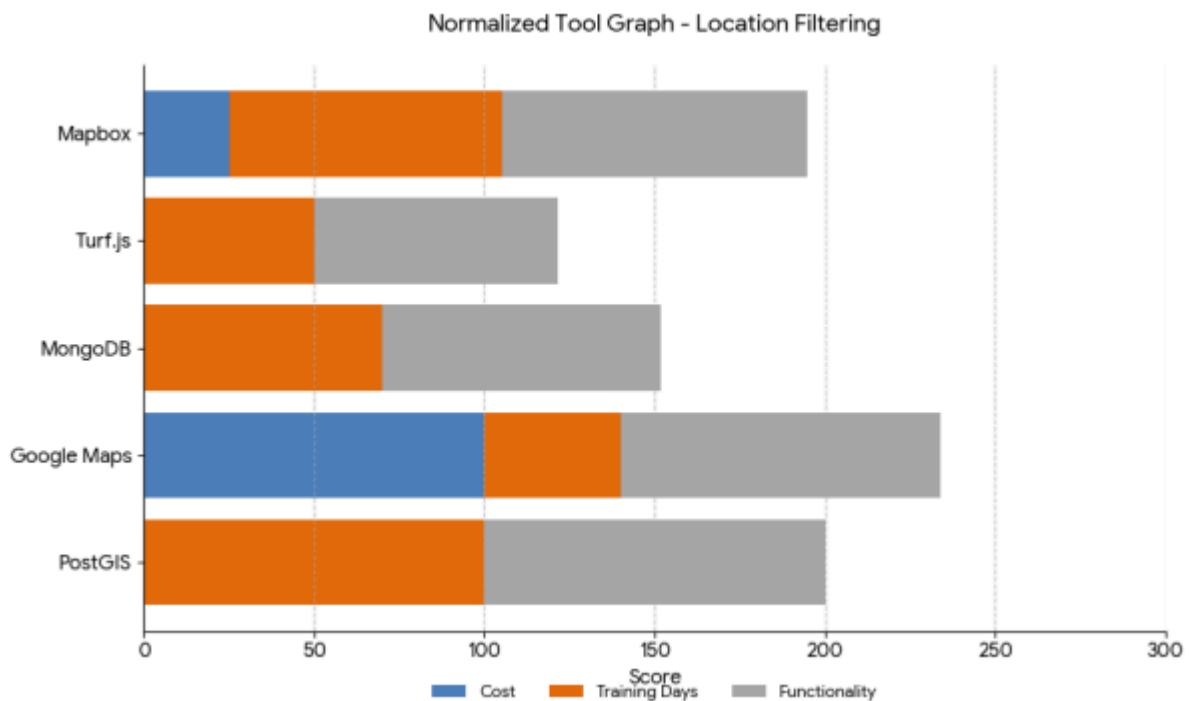
Task 3: Location Filtering:

Tool Cost/Training/Functionality Data

Tools	PostGIS	Google Maps API	MongoDB Geospatial	Turf.js	Mapbox SDK
Cost	Free	\$2,400/y	Free	Free	\$600/y
Training Days	10	4	7	5	8
Functionality	98	92	80	70	88

Normalized Cost/Training/Functionality Data

Tools	PostGIS	Google Maps API	MongoDB Geospatial	Turf.js	Mapbox SDK
Cost	0	100	0	0	25
Training Days	100	40	70	50	80
Functionality	100	93.88	81.63	71.43	89.90



We selected PostGIS as our location filtering tool because it offers the best balance between high-level functionality and zero cost. Since PostgreSQL was already selected for Task 2,

PostGIS provides enterprise-level geospatial capabilities without any additional financial burden. This choice ensures strong data management and security protocols necessary for our system requirements. While Google Maps and Mapbox offer high accuracy, they were ruled out due to their high annual costs. Furthermore, MongoDB and Turf.js were excluded to maintain a consistent and powerful relational database structure within our existing NestJS backend.

Task 4:Backend Framework :

Tool Cost/Training/Functionality Data

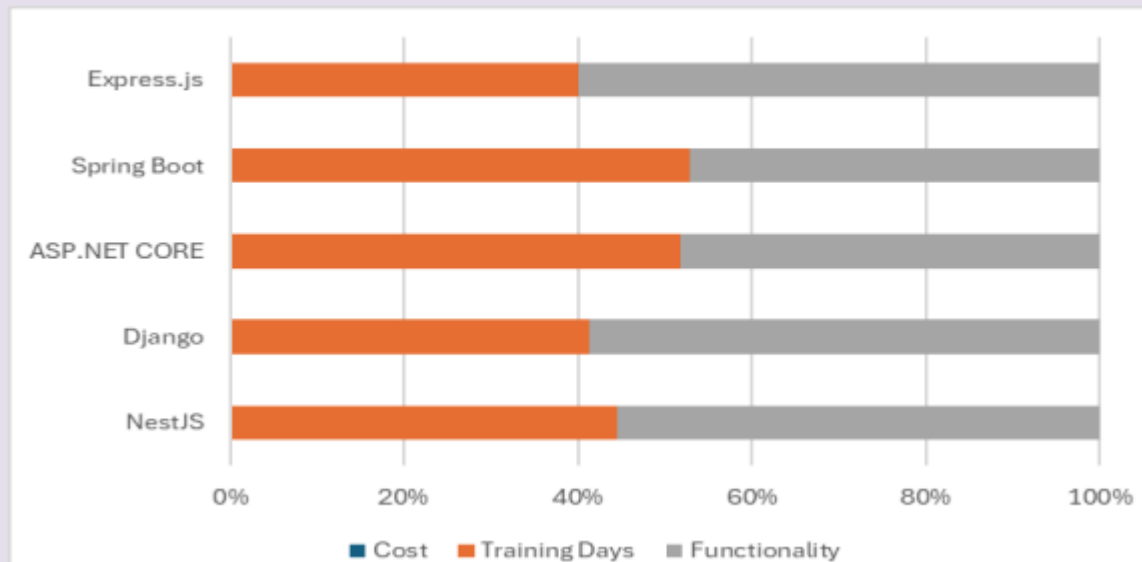
Tools	NestJS	Django	ASP.NET CORE	Spring Boot	Express.js
Cost	Free	Free	Free	Free	Free
Training Days	12	10	15	15	8
Functionality	95	90	88	85	75

Normalized Cost/Training/Functionality Data

Tools	NestJS	Django	ASP.NET CORE	Spring Boot	Express.js
Cost	0	0	0	0	0
Training Days	80	66.6	100	100	53.3

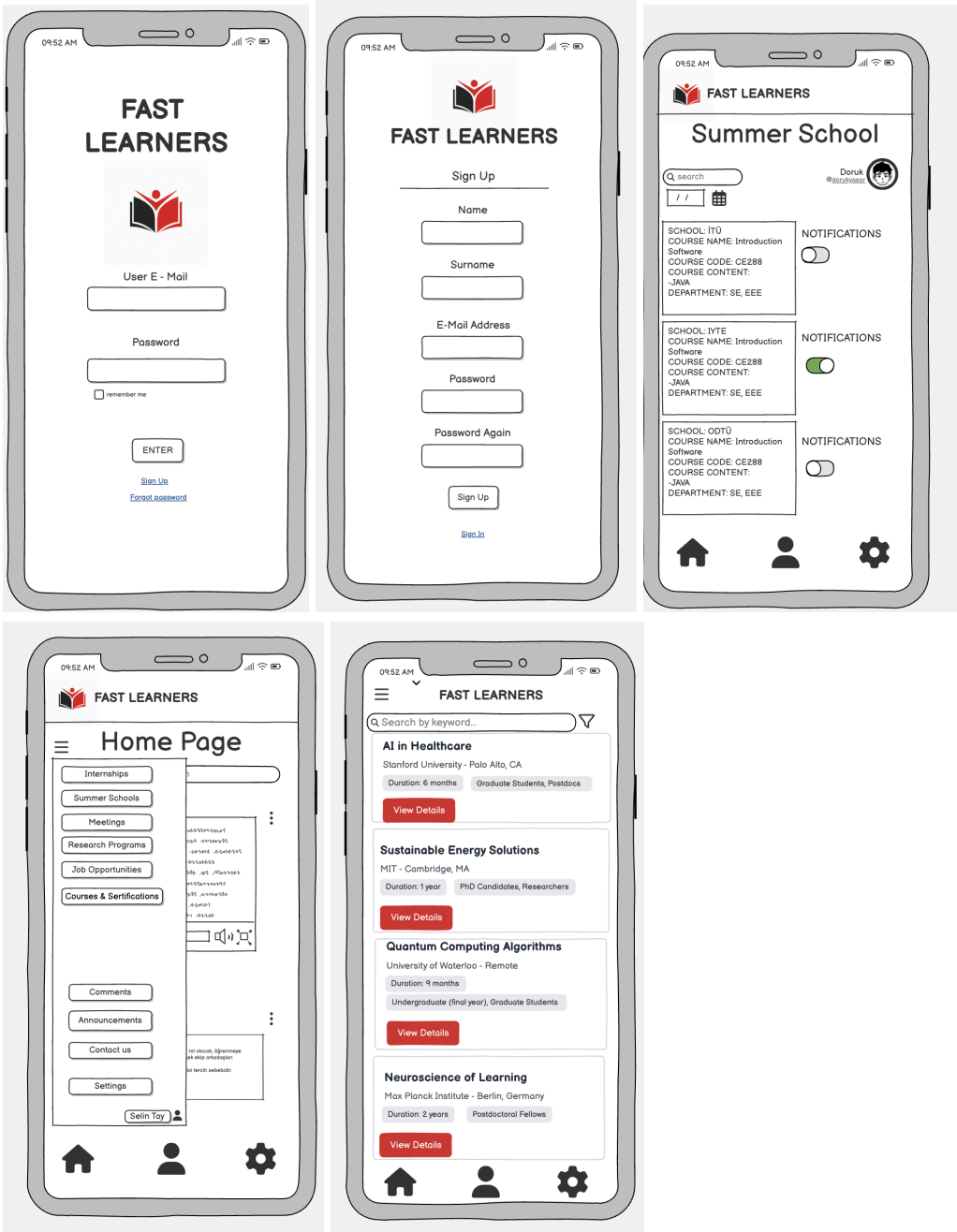
Functionality	100	94.7	92.6	89.4	78.9
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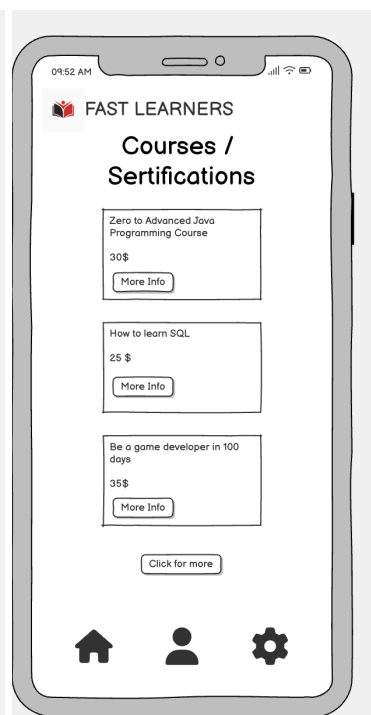
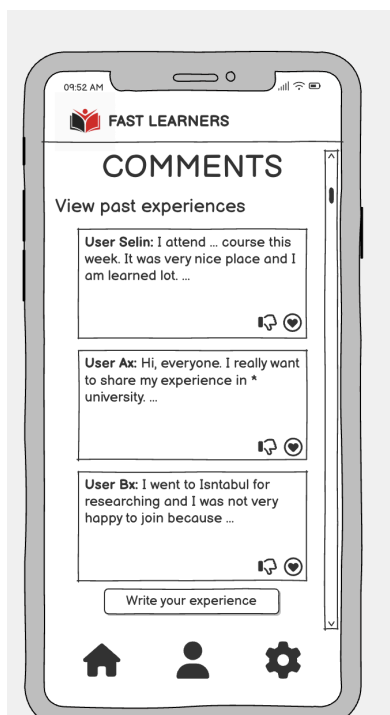
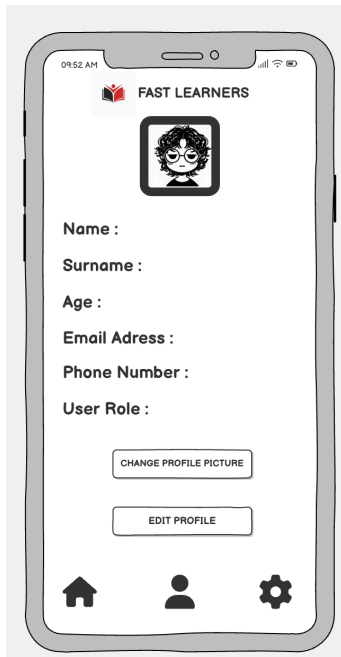
Normalized Tool Graph



We selected NestJS as our backend framework because it provides the best balance between modularity, functionality, and development speed for our project. Since backend frameworks are open-source, cost was not a deciding factor in our selection, and all options were considered equally in that aspect. NestJS stood out due to its modular structure, which allows us to organize features like user management, job listings, and filtering in a clean and scalable way while enabling team members to work in parallel. Although Django offers fast development and a strong admin panel, it is more monolithic and less flexible for a modular API-based system. ASP.NET Core and Spring Boot are powerful but require more learning time, which is not suitable for our limited schedule. Lastly, Express.js is simple and quick to use but lacks the structured architecture needed for maintaining a larger project. Therefore, NestJS was the most appropriate and balanced choice for our system.

8 - GRAPHICAL USER INTERFACES





9 - CONCLUSION

In conclusion, the “Centralized Platform for Educational and Career Opportunities” project aims to simplify access to educational and professional opportunities by bringing internships, job postings, scholarship programs, courses & certifications, summer schools, and research programs together in a single platform. Through features such as personalized recommendations, AI-based matching, filtering systems, notifications, and user reviews, the platform is designed to improve user experience and help students and young professionals make informed career decisions.

During the project analysis, both functional and non-functional requirements were identified to ensure the system’s usability, security, scalability, and accessibility. In addition, possible project risks such as inconsistent data sources, inaccurate AI recommendations, performance issues, notification failures, and security threats were evaluated through likelihood and impact analysis. Identifying these risks early allows the development team to create effective strategies to reduce their negative effects on the system.

Overall, this project has the potential to provide significant value for users by creating a reliable, user-friendly, and intelligent opportunity management platform. With proper planning, risk management, and continuous system improvement, the platform can become an efficient and sustainable solution for educational and career development.